

Purushotham Vadde

Senior Software Engineer | Backend & Distributed Systems

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PROFESSIONAL SUMMARY

Senior Software Engineer with 9+ years of experience designing distributed systems, cloud-native backend services, and large-scale data platforms. Proven expertise in Java, Spring Boot, Kafka, Spark, and AWS/Azure, delivering high-availability applications that process 10TB+ of data monthly and support enterprise-scale workloads. Passionate about building scalable systems, improving developer productivity, and leveraging AI to accelerate software development.

TECHNICAL SKILLS

Languages: Java, Python, Scala, SQL

Backend: Spring Boot, REST APIs, FastAPI

Distributed Systems: Microservices, Kafka, Event-Driven Architecture, Netflix Hollow

Data Engineering: Spark, Databricks, Airflow, Snowflake, Delta Lake, ETL/ELT

Cloud & Databases: AWS, Azure, MySQL, MongoDB, Redis, Cosmos DB

DevOps & Observability: Docker, Kubernetes, GitHub Actions, Argo CD, Liquibase, Grafana, Prometheus, Splunk

AI: Claude Code, LangGraph, LLMs, RAG, ChromaDB, Chainlit, Prompt Engineering

PROFESSIONAL EXPERIENCE

Senior Software Engineer

September 2021 – Present

Adobe Inc. | Denver, CO • Promoted from Software Engineer (September 2021 – December 2025)

- Led backend development for Adobe Journey Optimizer (AJO) B2B, designing scalable microservices and AI-assisted engineering workflows that reduced development effort by approximately 40%.
- Architected a distributed metadata platform using Netflix Hollow, enabling sub-millisecond lookups for real-time journey orchestration.
- Designed resilient event-processing services handling 500K+ Kafka events daily with 99.9% reliability through idempotent processing, retry orchestration, and automated recovery.
- Built Spark and Databricks data pipelines processing 10TB+ of data monthly, reducing data freshness from 24 hours to under 2 hours through incremental ingestion and automated validation.
- Delivered Adobe Experience Platform analytics pipelines using Spark, Airflow, and Azure Synapse, reducing reporting latency to under 30 minutes.
- Developed a real-time messaging platform supporting 10,000+ daily users with 99.9% uptime and sub-200 ms response times.
- Modernized CI/CD, database versioning, and observability across 50+ microservices, reducing deployment time from 4 hours to 20 minutes and improving production visibility.

Software Engineer

May 2015 – July 2019

Cognizant Technology Solutions | India

- Developed secure Python REST APIs for a Fortune 500 insurance platform, supporting role-based access control and PII-compliant data processing.
- Engineered PySpark ETL pipelines processing 5TB+ of data monthly on AWS EMR, improving performance by 2.5x through Spark optimization.

- Designed Amazon Redshift data warehouse solutions and automated ingestion pipelines, improving analytical query performance by 4x while reducing storage costs by 35%.

PROJECTS

JobHuntAI – AI-Powered Job Search Agent

2026 – Present

github.com/PurushothamVadde/job-hunt-agent | job-hunt-agent-sp7.onrender.com

- Built an AI-powered job search platform using LangGraph and Chainlit to automate job discovery, resume tailoring, and application workflows.
- Developed a RAG-based resume optimization pipeline using GPT-4o and ChromaDB to generate job-specific resumes from uploaded documents.
- Automated Workday applications using Playwright with human-in-the-loop approval for AI-assisted form completion.
- Implemented persistent user memory using SQLite and ChromaDB to personalize interactions across sessions.

EDUCATION

Doctoral Certificate in Business Administration

July 2024 – August 2025

Westcliff University

Master of Science, Data Science & Analytics

August 2019 – May 2021

University of Oklahoma